

[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

May-25-1099

MCA-6203 (Operating System)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from Sections A, B, C and D. Section E is compulsory.

SECTION - A

1. Define a Process Control Block (PCB). Explain its components in detail. How does context switching work? (12)
2. What is system boot? Explain the different steps involved in booting an operating system. (12)

SECTION - B

3. What are semaphores? Explain their role in synchronization with examples. (12)
4. Explain the communication mechanisms in client-server systems. Discuss the role of process synchronization in inter-process communication. (12)

SECTION - C

5. What is paging? Explain the working of paging with a suitable example. Discuss the concept of demand paging and its benefits. (12)
6. Explain the concepts of protection, access matrix, and access control. (12)

2

MCA-6203

SECTION - D

7. Discuss the standard Linux file system—ext3 and Virtual file system in Linux Operating system. (12)
8. Give in-depth analysis of Microsoft's design goals/principles for Windows operating system. (12)

SECTION - E (Compulsory)

9. Attempt ALL questions from this section. Each question carries 2 marks.

- (A) What is the benefit of multithreaded programming?
- (B) What do you understand by multiprogramming?
- (C) Why is inter-process communication important?
- (D) What do you understand by process synchronization?
- (E) Explain access control in context of operating system security.
- (F) What is process control block? (5×2=12)